Convergence of ADC(n)

These notes build on the Dyson vs. non-Dyson controversy, and assess how and whether the different ways to generate a systematic ADC known in the literature converge to the FCI limit.

I. THE λ-SCALED PROBLEM

Sector I are the $(N-1)$ configurations (2h1p, ...), sector II the $(N+1)$ ones (2p1h, ...). All results are properties of the Hamiltonian

$$\hat{H}(\lambda) = \hat{F} + \lambda(\hat{H} - \hat{F}), \tag{1}$$

where

$$\hat{F} = \sum_p \epsilon_p \hat{a}_p^\dagger \hat{a}_p \tag{2}$$

is the canonical HF Fock operator, built as dense matrices in the determinant spaces $\mathcal{H}_N$ and $\mathcal{H}_{N\pm1}$ of the systems in Table I. FCI gives the exact $\mathbf{\Sigma}(\omega;\lambda)$ at any λ; its Taylor coefficients follow from complex λ on a circle of radius $r = 0.3$ with $N_\lambda = 32$ points,[1]

$$\mathbf{\Sigma}^{(m)}(\omega) = \frac{1}{N_\lambda r^m} \sum_{j=0}^{N_\lambda - 1} \mathbf{\Sigma}(\omega; re^{2\pi \mathrm{i} j/N_\lambda}) e^{-2\pi \mathrm{i} m j/N_\lambda}. \tag{3}$$

The only perturbative input to ADC is the Rayleigh–Schrödinger (RS) series of the ground state and its energy,

$$|\Psi_0(\lambda)\rangle = \sum_{m\geq 0} \lambda^m \left|\Psi_0^{(m)}\right\rangle, \tag{4}$$

$$E_0(\lambda) = \sum_{m\geq 0} \lambda^m E_0^{(m)}. \tag{5}$$

The zeroth-order state is the HF determinant $|\Phi_0\rangle$, which is an eigenstate of $\hat{F}$,

$$\hat{F}\,|\Phi_0\rangle = E_0^{(0)}\,|\Phi_0\rangle\,, \tag{6}$$

with $E_0^{(0)}$ the sum of the occupied orbital energies. We use intermediate normalization, *i.e.* every correction is orthogonal to the HF determinant,

$$\left\langle \Phi_0 \middle| \Psi_0^{(m)} \right\rangle = \delta_{m0}. \tag{7}$$

Inserting the two series into the Schrödinger equation of $\hat{H}(\lambda)$ and collecting the terms of order λ^m gives the recursion

$$\left|\Psi_0^{(m)}\right\rangle = \hat{R}_0 \Big[(\hat{H} - \hat{F}) \left|\Psi_0^{(m-1)}\right\rangle - \sum_{l=1}^{m} E_0^{(l)} \left|\Psi_0^{(m-l)}\right\rangle \Big]. \tag{8}$$

Here $\hat{R}_0$ is the reduced resolvent of $\hat{F}$,

$$\hat{R}_0 = \hat{Q}\,(E_0^{(0)} - \hat{F})^{-1}\,\hat{Q}, \tag{9}$$

where

$$\hat{Q} = 1 - |\Phi_0\rangle\langle\Phi_0| \tag{10}$$

projects out the HF determinant. Because $\hat{F}$ is diagonal in the determinant basis, applying $\hat{R}_0$ amounts to dividing each determinant coefficient by the difference between $E_0^{(0)}$ and the orbital-energy sum of that determinant. The projector keeps the corrections orthogonal to $|\Phi_0\rangle$ and removes the one vanishing denominator. The energy corrections follow from projecting the order-l equation onto $|\Phi_0\rangle$,

$$E_0^{(l)} = \left\langle \Phi_0 \middle| \hat{H} - \hat{F} \middle| \Psi_0^{(l-1)} \right\rangle. \tag{11}$$

Since the matrix elements of Sec. II need the normalized state, every series is finally divided by the series of $\langle\Psi_0|\Psi_0\rangle$.

TABLE I. Systems (STO-3G). n_{occ}, n_{vir}: occupied and virtual spin orbitals; $k_{\max}$: last non-empty class; $n_{\mathrm{c}} = 2k_{\max}$: first order with the complete manifold; n_{op}: size of the complete manifold.

system	geometry (Å)	n_{occ}	n_{vir}	$k_{\max}$	n_{c}	n_{op}
H_4 chain	spacing 1.0	4	4	3	6	112
H_4 chain	spacing 1.6	4	4	3	6	112
LiH	1.6	4	8	4	8	1012
H_6 chain	spacing 1.0	6	6	5	10	1584

II. INTERMEDIATE-STATE BLOCKS

The excitation operators of the two sectors are the ionizing strings

$$\{\hat{r}_\nu^-\} = \{\hat{a}_b^\dagger \hat{a}_j \hat{a}_i,\ \hat{a}_c^\dagger \hat{a}_b^\dagger \hat{a}_k \hat{a}_j \hat{a}_i,\ \ldots\} \tag{12}$$

of sector I and the attaching strings

$$\{\hat{r}_\nu^+\} = \{\hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j,\ \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_c^\dagger \hat{a}_k \hat{a}_j,\ \ldots\} \tag{13}$$

of sector II. The ADC supermatrix is labelled by the single-character manifold

$$\{\hat{X}_J\} = \{\hat{a}_p^\dagger\} \cup \{\hat{r}_\nu^+\} \cup \{\hat{r}_\nu^{-\dagger}\}, \tag{14}$$

in which every operator adds one electron. It is ordered into classes k: $\hat{a}_p^\dagger$ is class 0; class k holds the $(k{+}1)$pkh strings of sector II and the adjoints of the $(k{+}1)$hkp strings of sector I. Each $\hat{X}_J$ becomes the vector pair $(\hat{X}_J\Psi_0, \hat{X}_J^\dagger\Psi_0)$ in $\mathcal{H}_{N+1} \oplus \mathcal{H}_{N-1}$, so that

$$\mathbb{S}_{IJ} = \langle \hat{X}_I\Psi_0 | \hat{X}_J\Psi_0\rangle + \langle \hat{X}_J^\dagger\Psi_0 | \hat{X}_I^\dagger\Psi_0\rangle \tag{15}$$

is the metric $\langle\{\hat{X}_I^\dagger, \hat{X}_J\}\rangle$ and

$$\mathbb{M}_{IJ} = \langle \hat{X}_I\Psi_0 | \hat{H} - E_0 | \hat{X}_J\Psi_0\rangle - \langle \hat{X}_J^\dagger\Psi_0 | \hat{H} - E_0 | \hat{X}_I^\dagger\Psi_0\rangle \tag{16}$$

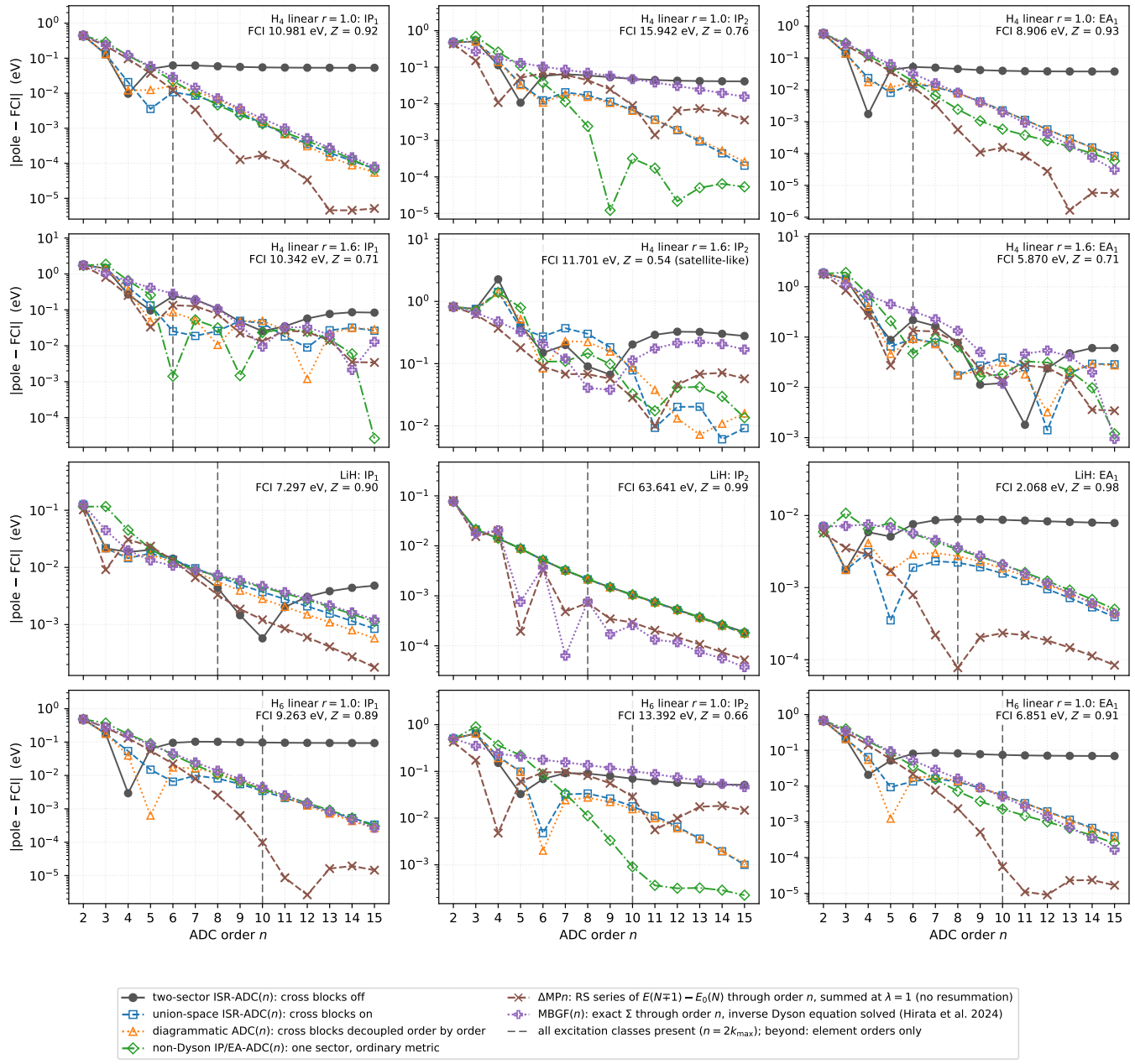


FIG. 1. Principal ADC(n) poles against FCI at $\lambda = 1$. Dashed line: $n = 2k_{\max}$, beyond which the manifold is complete (for LiH the non-Dyson IP sector is complete at $n = 6$).

is the superoperator $\langle\{\hat{X}_I^\dagger, [\hat{H}, \hat{X}_J]\}\rangle$. Inserting the RS series gives $\mathbb{S}^{(m)}$ and $\mathbb{M}^{(m)}$ exactly. The intermediate states[2–4] follow from Gram–Schmidt of class k against all lower classes and Löwdin orthonormalization within it, both done on the power series. The orthonormal supermatrix is

$$\tilde{\mathbb{M}} = \begin{pmatrix} \boldsymbol{\epsilon} + \boldsymbol{\Sigma}(\infty) & \mathbf{U}^{\mathrm{I}} & \mathbf{U}^{\mathrm{II}} \\ \mathbf{U}^{\mathrm{I}\dagger} & -\mathbf{K}^{\mathrm{I}} - \mathbf{C}^{\mathrm{I}} & \mathbf{X} \\ \mathbf{U}^{\mathrm{II}\dagger} & \mathbf{X}^\dagger & \mathbf{K}^{\mathrm{II}} + \mathbf{C}^{\mathrm{II}} \end{pmatrix}, \tag{17}$$

where $\mathbf{K}$ is the diagonal zeroth order and $\mathbf{X}$ the cross block between the sectors (its metric counterpart vanishes identically).

The inner projection[5] $\mathbf{G} = [\mathbb{S}(\omega\mathbb{S} - \mathbb{M})^{-1}\mathbb{S}]_{00}$ then has the Dyson form with

$$\boldsymbol{\Sigma}(\omega) = \boldsymbol{\Sigma}(\infty) + \mathbf{U}(\omega - \tilde{\mathbb{M}}_{\mathrm{cc}})^{-1}\mathbf{U}^\dagger, \tag{18}$$

$\tilde{\mathbb{M}}_{\mathrm{cc}}$ being all classes $k \geq 1$. Over the complete manifold this reproduces FCI to 10^{-13}. Folded order by order, the dynamic part $\boldsymbol{\Sigma}_{\mathrm{dyn}} = \boldsymbol{\Sigma} - \boldsymbol{\Sigma}(\infty)$ is

$$\boldsymbol{\Sigma}^{(m)}_{\mathrm{dyn}}(\omega) = \sum_{a+b+c=m} \mathbf{U}^{(a)}\, \mathbf{R}^{(b)}(\omega)\, \mathbf{U}^{(c)\dagger}, \tag{19}$$

where $\mathbf{R}^{(b)}$ is the order-b coefficient of $(\omega - \tilde{\mathbb{M}}_{\mathrm{cc}})^{-1}$. This is the exact Taylor reference; it agrees with the contour coefficients

to 5×10^{-13}. ADC(n)[6,7] keeps classes $k \le \lfloor n/2 \rfloor$ and block (k, l) through order $n - k - l$,

$$[\tilde{\mathbb{M}}_{\mathrm{ADC}(n)}]_{kl} = \sum_{m=0}^{n-k-l} \tilde{\mathbb{M}}_{kl}^{(m)}. \tag{20}$$

Its eigenvalues at $\lambda = 1$ are the poles; principal poles have orbital weight $Z > 0.3$. For $n \ge n_c$ (Table I) the manifold is complete and higher n only adds orders to its matrix elements.

III. CONSTRUCTIONS

All Dyson variants share $\mathbf{\Sigma}(\infty)$, $\mathbf{U}$ and the same-sector blocks exactly; they differ in $\mathbf{X}$.

TABLE II. The constructions of Fig. 1. Limit: complete manifold, all orders.

construction	$\mathbf{X}$	exact through	limit
two-sector ADC(n)	dropped	λ^3	not FCI
union ADC(n)	Eq. (20)	λ^n	FCI
diagrammatic ADC(n)	rotated	λ^n	FCI
non-Dyson ADC(n)	none	λ^n	FCI
ΔMPn	–	λ^n	RS series
MBGF(n)	–	$\mathbf{\Sigma}$ to λ^n	Dyson root

A. Union-space ISR-ADC(n)

Union-space ISR-ADC(n) keeps $\mathbf{X}$ under the rule of Eq. (20). The block between class k of sector I and class l of sector II starts at order

$$L_{kl} = \left\lceil \frac{k + l + 1}{2} \right\rceil, \tag{21}$$

so under the rule the block $\mathbf{X}_{11}$ enters at $n = 4$, $\mathbf{X}_{12}$ at $n = 5$, $\mathbf{X}_{22}$ and $\mathbf{X}_{13}$ at $n = 7$, $\mathbf{X}_{23}$ at $n = 8$ and $\mathbf{X}_{33}$ at $n = 10$. With these blocks the fold is exact through λ^n at every n tested ($n = 4$–9, to 10^{-12}).

B. Two-sector ISR-ADC(n) and the cross block

Two-sector ISR-ADC(n) orthonormalizes and folds each sector on its own, and thus yields the familiar form of the self-energy, organized by + and − branches,

$$\mathbf{\Sigma}(\omega) = \mathbf{\Sigma}(\infty) + \mathbf{U}^{\mathrm{II}}(\omega - \mathbf{K}^{\mathrm{II}} - \mathbf{C}^{\mathrm{II}})^{-1}\mathbf{U}^{\mathrm{II}\dagger} + \mathbf{U}^{\mathrm{I}}(\omega + \mathbf{K}^{\mathrm{I}} + \mathbf{C}^{\mathrm{I}})^{-1}\mathbf{U}^{\mathrm{I}\dagger}. \tag{22}$$

Its blocks are exactly the same-sector blocks of $\tilde{\mathbb{M}}$, so it is $\tilde{\mathbb{M}}$ with $\mathbf{X}$ deleted. The cross block vanishes at the two lowest orders,

$$\mathbf{X}^{(0)} = \mathbf{X}^{(1)} = 0, \tag{23}$$

but not at second, where it is a contraction of the first-order doubles with the two-electron integrals. Expanding the union fold in $\mathbf{X}$, the leading term the two-sector construction misses is

$$\Delta\mathbf{\Sigma}^{(4)} = \mathbf{U}^{\mathrm{I}(1)}(\omega + \mathbf{K}_1^{\mathrm{I}})^{-1}\mathbf{X}_{11}^{(2)}(\omega - \mathbf{K}_1^{\mathrm{II}})^{-1}\mathbf{U}^{\mathrm{II}(1)\dagger} + \mathrm{h.c.}, \tag{24}$$

which matches the λ^4 difference of the two folds to 5×10^{-18}. Two-sector ISR-ADC(n) is therefore exact only through λ^3, for every n.

As identified by Tarantelli and Cederbaum,[5] the defect is fundamental. The complete-manifold limit of the Dyson-ISR construction is not FCI as can be seen clearly in Fig. 1. Tarantelli and Cederbaum write the Dyson equation as a configuration-interaction problem in $\mathcal{H}_{N+1} \oplus \mathcal{H}_{N-1}$, with a composite Hamiltonian that is $\mathbb{M}$ of Eq. (16). The block that couples the $(N + 1)$ and $(N - 1)$ configurations vanishes in zeroth and first order but not in second, and they conclude that this coupling "prevents the possibility of reproducing analytically the fourth-order ADC scheme", since ADC assumes that the self-energy splits into independent $(N + 1)$ and $(N - 1)$ parts. They remove the coupling by a further unitary transformation that block-diagonalizes the configuration space. This transformation can only be built order by order, and with an exactly solvable two-electron example they show that no transformation built from the ground state alone achieves the decoupling. The two-sector ISR is built from the ground state alone and is decoupled by construction.

C. Diagrammatic ADC(n)

Diagrammatic ADC(n) removes $\mathbf{X}$ by the Tarantelli–Cederbaum rotation[5] $\mathbf{W} = e^{\mathbf{A}}$, with $\mathbf{A}$ antisymmetric in the I–II blocks and

$$A_{\mu\nu}^{(m)} = \frac{R_{\mu\nu}^{(m)}}{K_\nu - K_\mu}, \tag{25}$$

$\mathbf{R}^{(m)}$ being the I–II part of $[\mathbf{W}^\intercal \tilde{\mathbb{M}} \mathbf{W}]^{(m)}$ from $\mathbf{A}^{(<m)}$; Eq. (20) is applied to the rotated blocks. At third order this moves $\mathbf{X}^{(2)}$ into the couplings,

$$\delta U_{p\nu}^{\mathrm{II}(3)} = \sum_{\mu \in \mathrm{I}_1} \frac{U_{p\mu}^{\mathrm{I}(1)} X_{\mu\nu}^{(2)}}{K_\nu - K_\mu}, \tag{26}$$

where $K_\nu - K_\mu$ is a 3p3h energy; these rows reproduce the diagrammatic ADC(4) rows[7] to 10^{-18}.

It is a remarkable connection that the usual diagrammatic construction of ADC (hence the name) by Schirmer, Cederbaum, and Walter[7] that takes diagrams up to order n and enforces the correct analytical form of the exact self-energy is equivalent to the blockwise decoupling of Tarantelli and Cederbaum[5].

D. Non-Dyson ADC(n)

Non-Dyson ADC(n)[8,9] builds the ISR from $\hat{r}_\nu^- \Psi_0$ in $\mathcal{H}_{N-1}$ (or $\hat{r}_\nu^+ \Psi_0$ in $\mathcal{H}_{N+1}$) with the ordinary overlap and the same rule; its eigenvalues are the IPs (EAs).

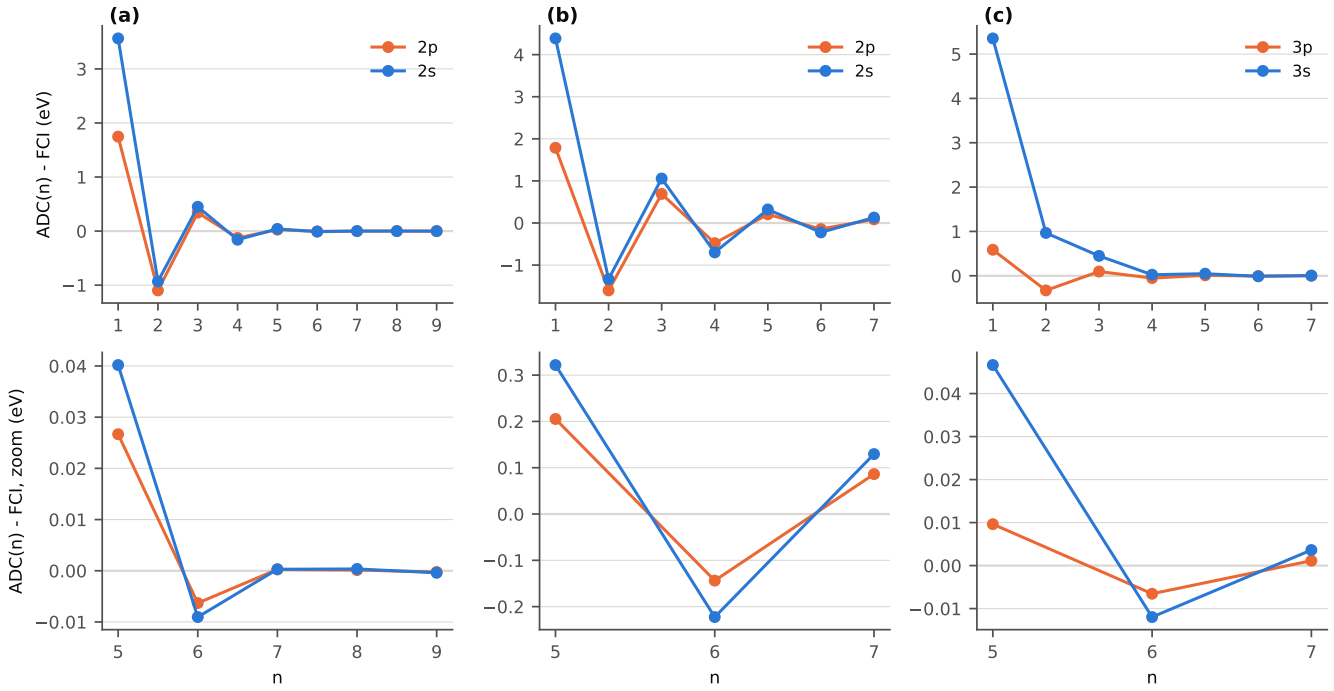


FIG. 2. Non-Dyson ADC(n) against FCI. (a) Ne/cc-pVDZ, 1s frozen; (b) Ne/aug-cc-pVDZ, 1s frozen; (c) Ar/aug-cc-pVDZ, Ne core (1s2s2p) frozen. Top: $n = 1$ (Koopmans) to the last order; bottom: $n \geq 5$, each with its own vertical range.

E. Static self-energy

Dyson ADC(n) evaluates $\Sigma(\infty)$ with the static MP($n-1$) density matrix. In the non-Dyson case, IP-ADC(n) is, for a given orthonormalization of the intermediate states, the unique Hermitian, order-n-consistent way to embed the EA sector statically into the $(N-1)$-electron space. At $n = 2$ this is equivalent to folding the EA sector statically, at the orbital energies, into the 1-particle sector; at $n = 3$ the two give the same IPs through λ^3. Starting from ADC(4), the consistent ISR construction and the static fold of the EA sector into the 1-particle sector differ: the naive static fold introduces an error at order λ^4.

F. Perturbative references

ΔMPn is the RS series of $E_{N-1}(\lambda) - E_0(\lambda)$ for the Koopmans-connected state, summed through order n. MBGF(n)[10] takes the exact $\Sigma^{[n]} = \sum_{m \leq n} \Sigma^{(m)}$ and solves

$$\omega = e_p(\omega), \tag{27}$$

with e_p the eigenvalue of $\epsilon + \Sigma^{[n]}(\omega)$ of largest weight on p, taking the root of largest residue. The pole Taylor series of union, diagrammatic and non-Dyson ADC(n) equal the exact one through λ^n to 10^{-17} ($n \leq 15$); the two-sector pole deviates from λ^4 on.

IV. NE AND AR BEYOND ADC(5)

Fig. 2 follows the non-Dyson ADC(n) series of the Ne and Ar valence ionizations in correlation-consistent bases to high order and compares against the FCI IP of the same block. Fig. 2(a) shows the result for Ne in the cc-pVDZ basis set, and a clear oscillatory convergence can be observed. In line with the observation of Olsen *et al.* [11], using an augmented basis set as in Fig. 2(b) amplifies the oscillations. The pattern is similar for Ar, but the oscillations are milder.

V. ACCURACY BENCHMARK OF ADC(2)–(5)

We compare Dyson diagrammatic, Dyson two-sector and non-Dyson ADC(2)–(5) against the reference results from the benchmark set of Marie and Loos [12] against their TBEs at the sCI level of theory, as well as against their CC values, from CC2 to CCSDT. All calculations use the same frozen-core approximation as Marie and Loos [12]. The high computational cost and storage requirements of ADC(5) prevented us from solving for all states of CH_2O and CO_2.

A. Basis-set dependence

Fig. 3 gives the change of the MAD from aug-cc-pVDZ to aug-cc-pVTZ, whereby the result in each basis is compared against its own sCI reference. The comparison is performed

on the 19 states of 15 molecules for which the two-sector ADC(5) is available in both bases. Up to ADC(4) the larger basis lowers the MAD by 15 to 45 meV, but the choice of basis matters less and less with increasing order. At ADC(5) the two-sector MAD rises by only 5 meV. The basis therefore does not change the convergence pattern. The rest of this section uses aug-cc-pVDZ.

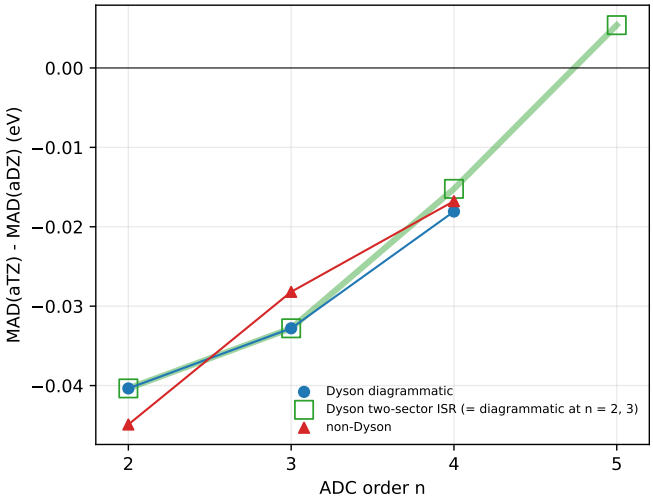


FIG. 3. MAD(aug-cc-pVTZ) − MAD(aug-cc-pVDZ) against n, each basis against its own sCI, for the 19 states of 15 molecules with two-sector ADC(5) in both bases, frozen core. Dyson two-sector ADC(n) equals the diagrammatic one at $n = 2, 3$.

B. Convergence with the order

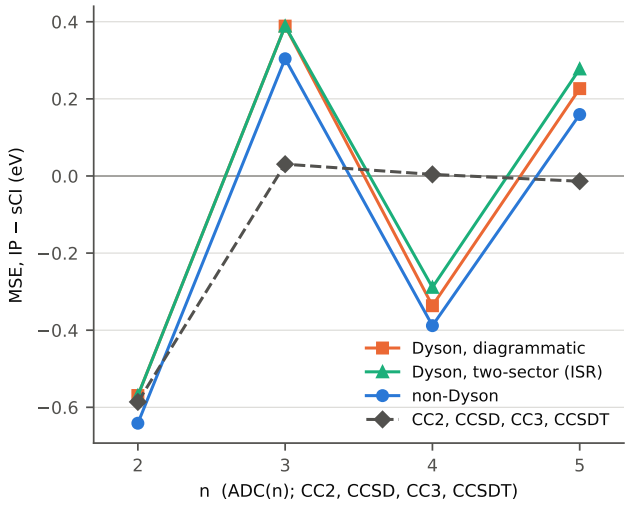


FIG. 4. Mean signed error IP − sCI against n over the 50 IPs of 21 molecules in the set by Marie and Loos [12] that have every column of Table III.

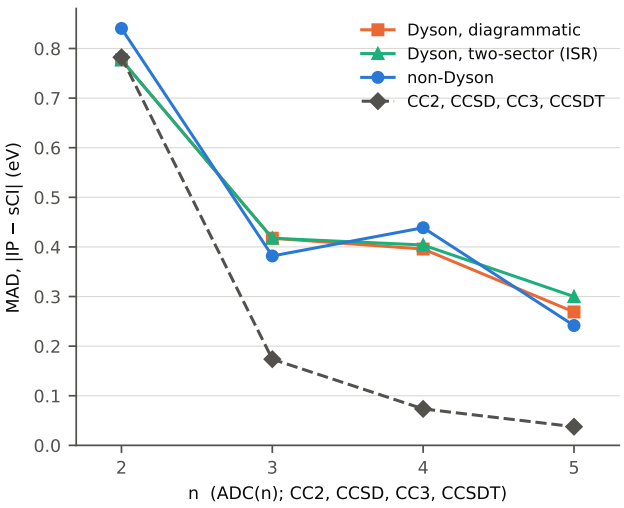


FIG. 5. Mean absolute error, as Fig. 4.

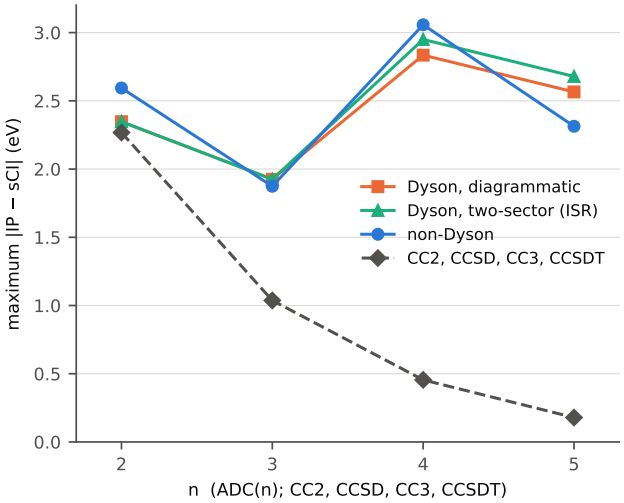


FIG. 6. Maximum absolute error, as Fig. 4.

Figs. 4–6 show the mean signed, mean absolute and maximum absolute error of the 50 states against n at aug-cc-pVDZ. As a consequence of the oscillating convergence pattern also observed for Neon, the ADC error alternates in sign. It should be noted that this pattern is observed on average also for the molecules in the set of Marie and Loos [12], but it does not always follow the oscillating trend we observed for Neon. The MAD halves from ADC(2) to ADC(3), but hardly changes from ADC(3) to ADC(4). At ADC(5), it reaches values of 0.27 (diagrammatic), 0.30 (two-sector) and 0.24 eV (non-Dyson). This is to be contrasted with CCSDT, which comes with a MAD of only 40 meV. This much better performance also comes at a lower practical computational cost, as ADC(5) requires constructing 8-index tensors. The maximum error is observed for BeO, whose two IPs are off by about 2 to 3 eV at every order. In contrast, CCSDT is within 0.14 eV for both

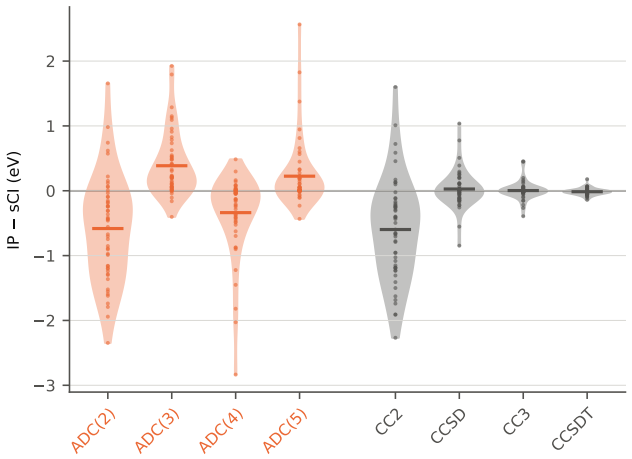


FIG. 7. Distribution of the IP errors against sCI of Dyson diagrammatic ADC(2)–(5) and of CC2, CCSD, CC3, CCSDT over the 52 states of 22 molecules (Table III) for which diagrammatic ADC(5) values exist. All values are in eV.

states of BeO.

It should be remarked that for many, typically weakly correlated systems, ADC(5) performs very well (see Figs. 8–11 for all individual cases). But even in these cases, no advantage over CC can be observed. However, as seen in Fig. 7, after ADC(3), the spread of errors does not noticeably reduce further.

C. Individual states

Figs. 8–11 show the error of every one of the 58 states against n for all three constructions and CC, and Table III lists the numbers. For most states the ADC error alternates in sign with n and does not decrease monotonically.

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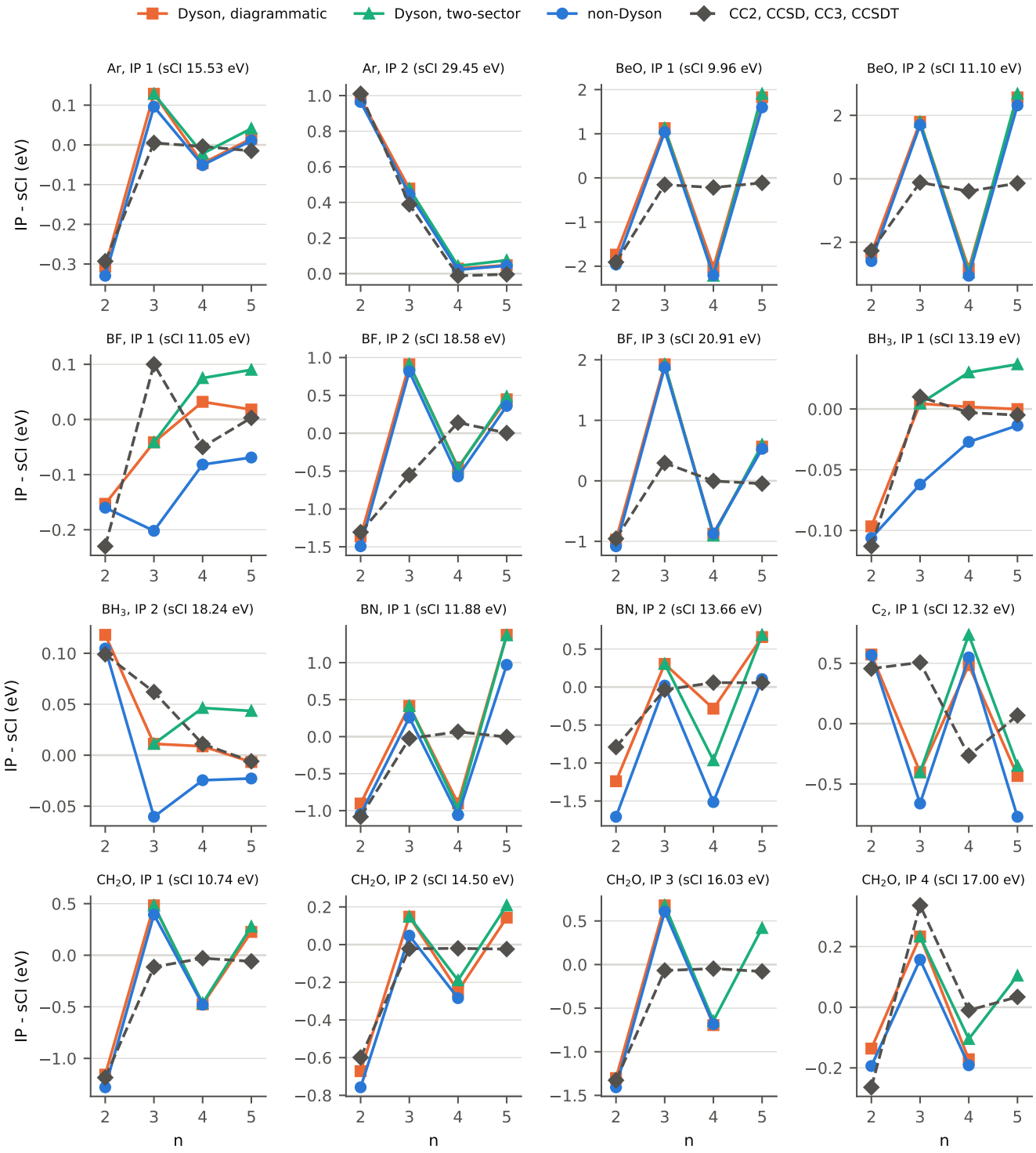


FIG. 8. Error IP − sCI against order n for all molecules in the set by Marie and Loos [12]. All calculations are frozen core and in the aug-cc-pVDZ basis set (1/4).

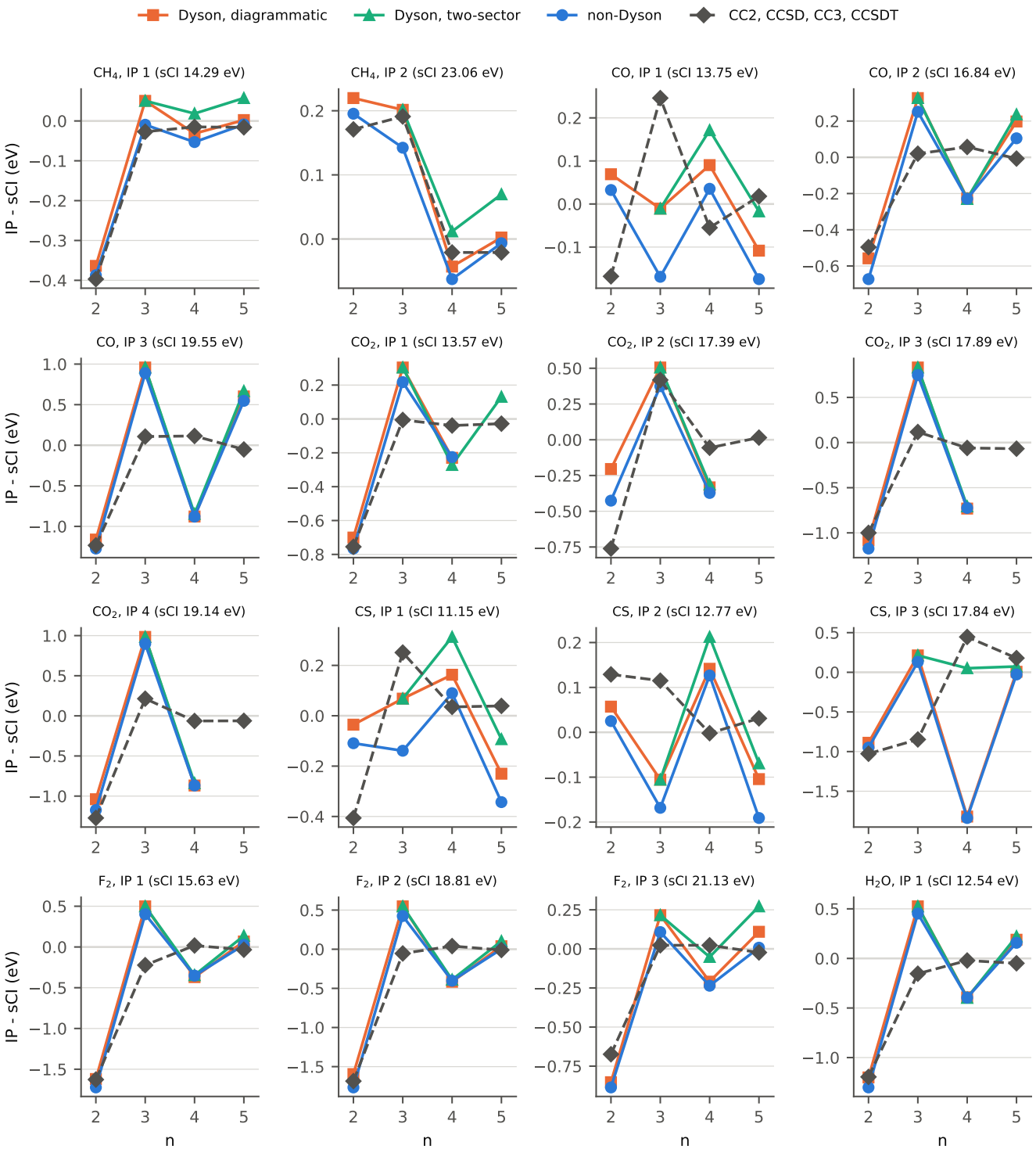


FIG. 9. As Fig. 8 (2/4).

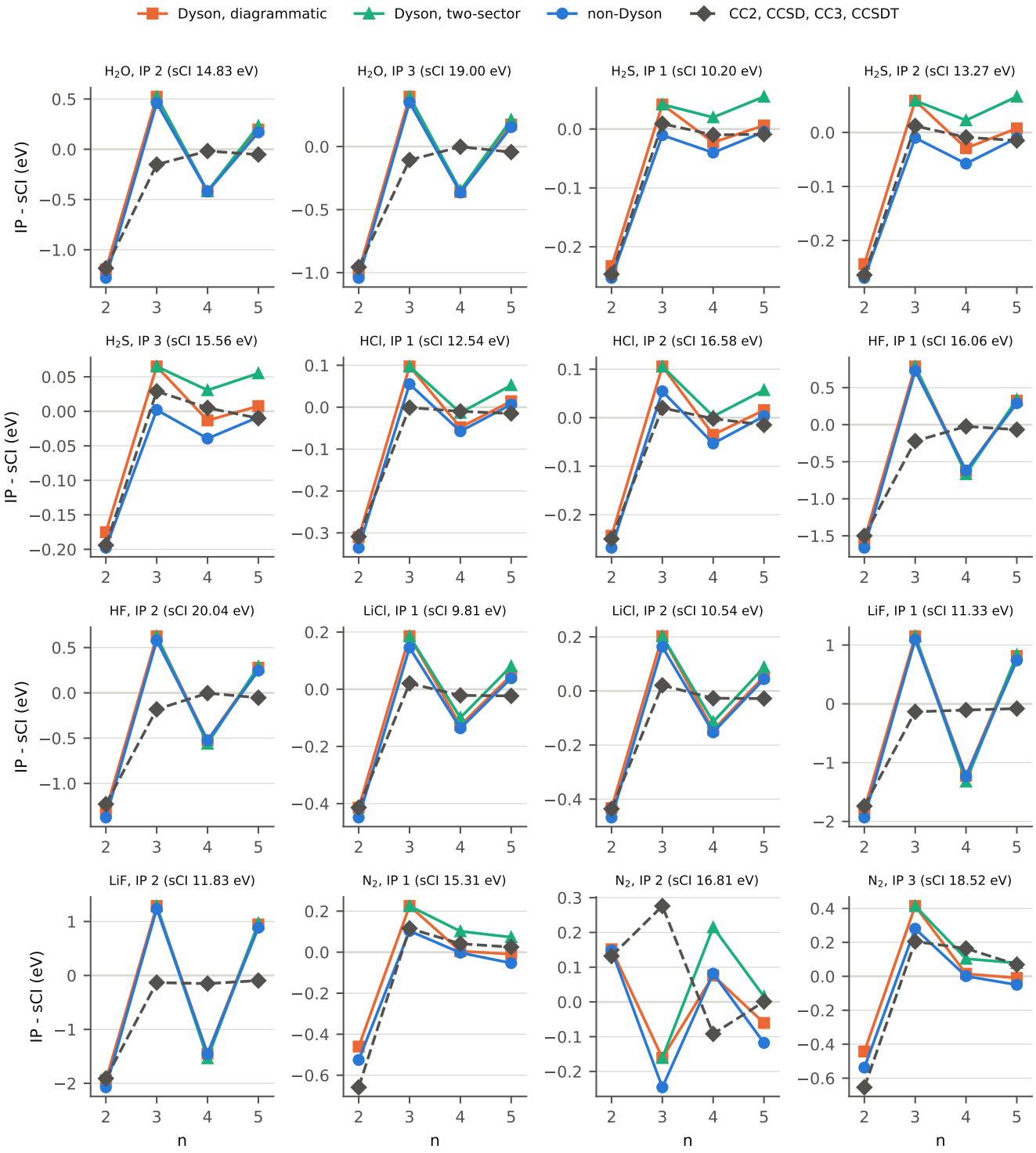


FIG. 10. As Fig. 8 (3/4).

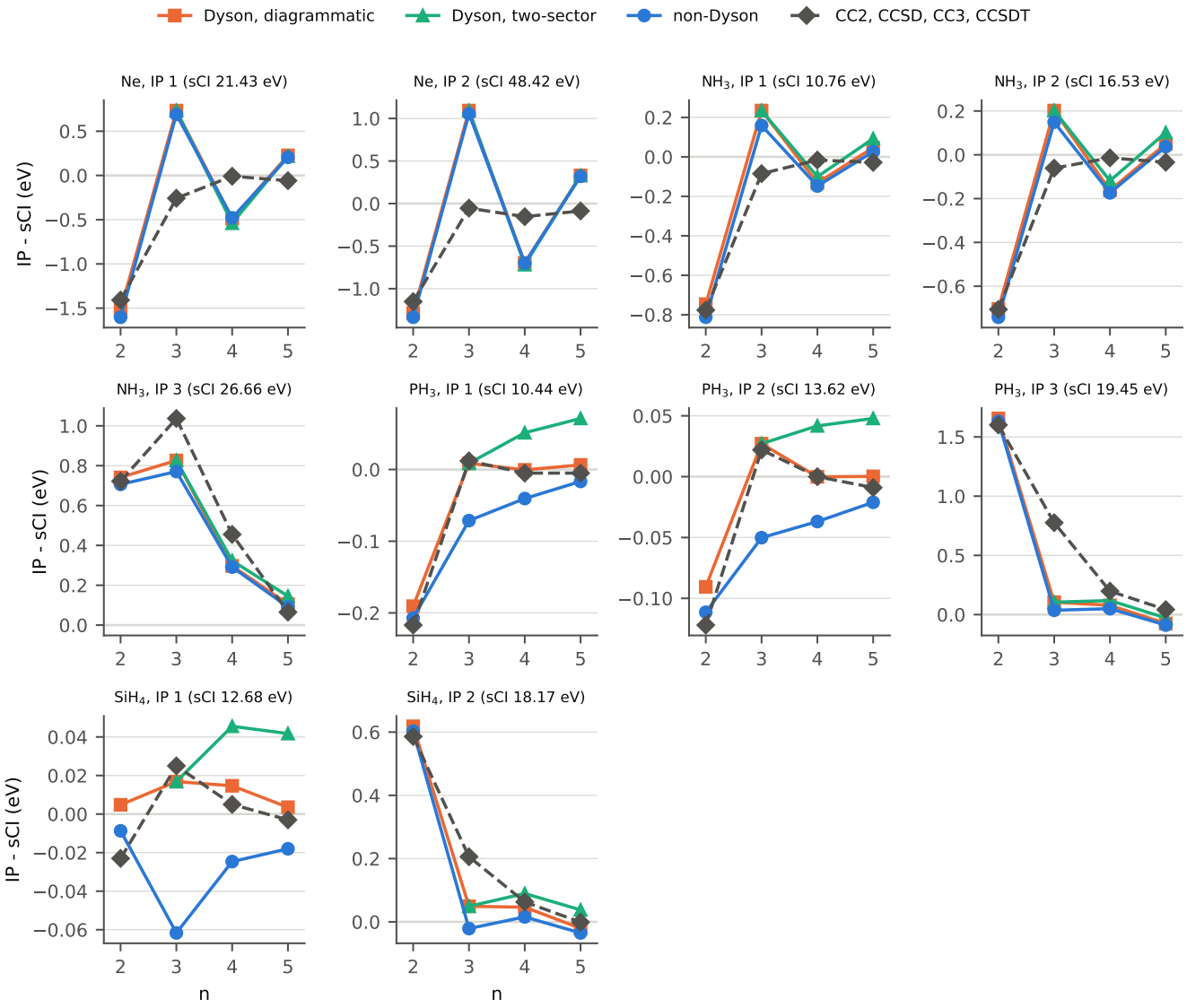


FIG. 11. As Fig. 8 (4/4).

TABLE III. Ionization energies of the SupInf set[12] at aug-cc-pVDZ, frozen core ADC(n): sCI (eV) of Ref. 12 and the error Δ = method − sCI (eV) of Dyson diagrammatic ADC(2)–(5), Dyson two-sector ADC(4) and ADC(5) (identical to diagrammatic below fourth order), non-Dyson ADC(2)–(5) and CC2, CCSD, CC3, CCSDT (Ref. 12). MSE, MAD and maximum |Δ| over the 50 states of 21 molecules that have every column (the state set of Figs. 4–6), all and valence only.

			Dyson, diagrammatic				Dyson, two-sector		non-Dyson				coupled cluster			
Molecule	State	sCI	(2)	(3)	(4)	(5)	(4)	(5)	(2)	(3)	(4)	(5)	CC2	CCSD	CC3	CCSDT
Ar	1	15.53	−0.31	+0.13	−0.05	+0.02	−0.02	+0.04	−0.33	+0.10	−0.05	+0.01	−0.29	+0.01	+0.00	−0.01
	2[b]	29.45	+0.98	+0.48	+0.03	+0.05	+0.04	+0.07	+0.96	+0.45	+0.02	+0.04	+1.01	+0.39	−0.01	+0.00
BeO	1	9.96	−1.74	+1.13	−2.03	+1.83	−2.22	+1.90	−1.96	+1.04	−2.21	+1.60	−1.91	−0.15	−0.22	−0.11
	2	11.10	−2.35	+1.79	−2.83	+2.57	−2.95	+2.68	−2.59	+1.70	−3.06	+2.31	−2.27	−0.12	−0.39	−0.14
BF	1	11.05	−0.15	−0.04	+0.03	+0.02	+0.08	+0.09	−0.16	−0.20	−0.08	−0.07	−0.23	+0.10	−0.05	+0.00
	2	18.58	−1.36	+0.91	−0.45	+0.45	−0.46	+0.49	−1.49	+0.82	−0.57	+0.36	−1.31	−0.55	+0.14	+0.00
	3	20.91	−0.97	+1.93	−0.87	+0.57	−0.90	+0.60	−1.08	+1.87	−0.87	+0.53	−0.96	+0.29	+0.00	−0.05
BH_3	1	13.19	−0.10	+0.00	+0.00	+0.00	+0.03	+0.04	−0.11	−0.06	−0.03	−0.01	−0.11	+0.01	+0.00	−0.01
	2	18.24	+0.12	+0.01	+0.01	−0.01	+0.05	+0.04	+0.10	−0.06	−0.02	−0.02	+0.10	+0.06	+0.01	−0.01
BN	1[a]	11.88	−0.90	+0.42	−0.90	+1.38	−0.98	+1.37	−1.05	+0.26	−1.06	+0.97	−1.08	−0.03	+0.07	+0.00
	2[a]	13.66	−1.24	+0.30	−0.28	+0.66	−0.96	+0.69	−1.71	+0.02	−1.51	+0.10	−0.79	−0.04	+0.06	+0.06
C_2	1[a]	12.32	+0.57	−0.40	+0.49	−0.43	+0.74	−0.35	+0.57	−0.66	+0.55	−0.77	+0.46	+0.51	−0.27	+0.07
CH_2O	1	10.74	−1.16	+0.48	−0.48	+0.23	−0.47	+0.28	−1.28	+0.39	−0.48	—	−1.19	−0.11	−0.03	−0.06
	2	14.50	−0.67	+0.15	−0.25	+0.14	−0.19	+0.21	−0.76	+0.05	−0.28	—	−0.60	−0.02	−0.02	−0.02
	3	16.03	−1.30	+0.68	−0.69	—	−0.65	+0.42	−1.41	+0.61	−0.68	—	−1.33	−0.07	−0.05	−0.08
	4	17.00	−0.14	+0.23	−0.17	—	−0.10	+0.10	−0.19	+0.16	−0.19	—	−0.26	+0.34	−0.01	+0.03
CH_4	1	14.29	−0.36	+0.05	−0.03	+0.00	+0.02	+0.06	−0.39	−0.01	−0.05	−0.01	−0.40	−0.03	−0.01	−0.02
	2[b]	23.06	+0.22	+0.20	−0.04	+0.00	+0.01	+0.07	+0.20	+0.14	−0.06	−0.01	+0.17	+0.19	−0.02	−0.02
CO	1	13.75	+0.07	−0.01	+0.09	−0.11	+0.17	−0.02	+0.03	−0.17	+0.04	−0.17	−0.17	+0.25	−0.06	+0.02
	2	16.84	−0.56	+0.33	−0.23	+0.20	−0.23	+0.24	−0.67	+0.25	−0.23	+0.10	−0.50	+0.02	+0.06	−0.01
	3	19.55	−1.16	+0.96	−0.88	+0.60	−0.85	+0.67	−1.27	+0.89	−0.88	+0.55	−1.23	+0.11	+0.11	−0.05
CO_2	1	13.57	−0.70	+0.30	−0.23	—	−0.27	+0.13	−0.77	+0.22	−0.23	—	−0.76	−0.01	−0.04	−0.03
	2	17.39	−0.20	+0.51	−0.33	—	−0.31	—	−0.43	+0.37	−0.37	—	−0.76	+0.42	−0.06	+0.02
	3	17.89	−1.08	+0.83	−0.73	—	−0.71	—	−1.17	+0.75	−0.72	—	−1.00	+0.12	−0.06	−0.07
	4	19.14	−1.04	+0.98	−0.87	—	−0.84	—	−1.17	+0.90	−0.87	—	−1.27	+0.21	−0.06	−0.06
CS	1	11.15	−0.04	+0.07	+0.16	−0.23	+0.31	−0.09	−0.11	−0.14	+0.09	−0.34	−0.41	+0.25	+0.04	+0.04
	2	12.77	+0.06	−0.11	+0.14	−0.10	+0.21	−0.07	+0.02	−0.17	+0.13	−0.19	+0.13	+0.12	+0.00	+0.03
	3	17.84	−0.89	+0.21	−1.82	+0.01	+0.05	+0.07	−0.95	+0.13	−1.84	−0.03	−1.03	−0.85	+0.45	+0.18
F_2	1	15.63	−1.62	+0.50	−0.37	+0.07	−0.35	+0.14	−1.72	+0.40	−0.35	+0.03	−1.63	−0.22	+0.02	−0.03
	2	18.81	−1.60	+0.55	−0.42	+0.04	−0.39	+0.10	−1.77	+0.42	−0.40	+0.00	−1.69	−0.05	+0.04	−0.01
	3	21.13	−0.85	+0.22	−0.21	+0.11	−0.05	+0.27	−0.89	+0.11	−0.24	+0.01	−0.67	+0.02	+0.02	−0.02
H_2O	1	12.54	−1.20	+0.53	−0.39	+0.19	−0.40	+0.22	−1.30	+0.45	−0.40	+0.16	−1.20	−0.15	−0.02	−0.05
	2	14.83	−1.20	+0.52	−0.42	+0.20	−0.42	+0.23	−1.28	+0.46	−0.42	+0.17	−1.18	−0.15	−0.02	−0.05
	3	19.00	−0.99	+0.40	−0.36	+0.17	−0.35	+0.21	−1.04	+0.35	−0.37	+0.15	−0.96	−0.11	+0.00	−0.04
H_2S	1	10.20	−0.23	+0.04	−0.02	+0.01	+0.02	+0.05	−0.25	−0.01	−0.04	+0.00	−0.25	+0.01	−0.01	−0.01
	2	13.27	−0.24	+0.06	−0.03	+0.01	+0.02	+0.07	−0.27	−0.01	−0.06	−0.01	−0.26	+0.01	−0.01	−0.01
	3	15.56	−0.18	+0.07	−0.01	+0.01	+0.03	+0.06	−0.20	+0.00	−0.04	−0.01	−0.19	+0.03	+0.01	−0.01
HCl	1	12.54	−0.31	+0.10	−0.05	+0.01	−0.01	+0.05	−0.34	+0.06	−0.06	+0.01	−0.31	+0.00	−0.01	−0.01
	2	16.58	−0.24	+0.11	−0.04	+0.02	+0.00	+0.06	−0.27	+0.05	−0.05	+0.00	−0.25	+0.02	+0.00	−0.01
HF	1	16.06	−1.55	+0.79	−0.63	+0.32	−0.67	+0.34	−1.66	+0.73	−0.62	+0.29	−1.50	−0.22	−0.02	−0.07
	2	20.04	−1.30	+0.63	−0.52	+0.28	−0.56	+0.29	−1.38	+0.58	−0.52	+0.25	−1.23	−0.18	+0.00	−0.05
LiCl	1	9.81	−0.41	+0.19	−0.13	+0.05	−0.10	+0.08	−0.45	+0.15	−0.14	+0.04	−0.41	+0.02	−0.02	−0.02
	2	10.54	−0.43	+0.20	−0.14	+0.05	−0.11	+0.09	−0.47	+0.16	−0.15	+0.04	−0.44	+0.02	−0.03	−0.03
LiF	1	11.33	−1.79	+1.15	−1.22	+0.81	−1.32	+0.83	−1.93	+1.09	−1.24	+0.74	−1.74	−0.14	−0.11	−0.08
	2	11.83	−1.94	+1.29	−1.45	+0.95	−1.54	+0.97	−2.07	+1.23	−1.45	+0.88	−1.91	−0.13	−0.15	−0.10
N_2	1	15.31	−0.46	+0.23	+0.00	−0.01	+0.10	+0.07	−0.53	+0.10	+0.00	−0.05	−0.66	+0.12	+0.04	+0.03
	2	16.81	+0.15	−0.16	+0.08	−0.06	+0.21	+0.02	+0.15	−0.25	+0.08	−0.12	+0.13	+0.28	−0.09	+0.00
	3	18.52	−0.44	+0.41	+0.01	−0.01	+0.10	+0.08	−0.54	+0.28	+0.00	−0.05	−0.65	+0.20	+0.16	+0.07
Ne	1	21.43	−1.52	+0.73	−0.49	+0.23	−0.54	+0.22	−1.60	+0.69	−0.48	+0.20	−1.41	−0.26	−0.01	−0.06
	2[b]	48.42	−1.29	+1.09	−0.70	+0.33	−0.72	+0.33	−1.34	+1.06	−0.70	+0.32	−1.15	−0.05	−0.15	−0.09
NH_3	1	10.76	−0.75	+0.23	−0.13	+0.05	−0.10	+0.09	−0.81	+0.16	−0.15	+0.03	−0.78	−0.09	−0.02	−0.03
	2	16.53	−0.70	+0.20	−0.16	+0.05	−0.12	+0.10	−0.74	+0.15	−0.17	+0.04	−0.71	−0.06	−0.01	−0.03
	3[b]	26.66	+0.74	+0.83	+0.30	+0.11	+0.32	+0.15	+0.71	+0.77	+0.29	+0.09	+0.72	+1.04	+0.46	+0.07
PH_3	1	10.44	−0.19	+0.01	+0.00	+0.01	+0.05	+0.07	−0.21	−0.07	−0.04	−0.02	−0.22	+0.01	−0.01	−0.01
	2	13.62	−0.09	+0.03	+0.00	+0.00	+0.04	+0.05	−0.11	−0.05	−0.04	−0.02	−0.12	+0.02	+0.00	−0.01
	3[b]	19.45	+1.66	+0.10	+0.08	−0.08	+0.12	−0.03	+1.63	+0.04	+0.05	−0.09	+1.60	+0.78	+0.20	+0.04
SiH_4	1	12.68	+0.00	+0.02	+0.01	+0.00	+0.05	+0.04	−0.01	−0.06	−0.02	−0.02	−0.02	+0.03	+0.01	+0.00
	2	18.17	+0.62	+0.05	+0.05	−0.02	+0.09	+0.04	+0.60	−0.02	+0.02	−0.04	+0.59	+0.21	+0.06	+0.00
MSE, all states			−0.57	+0.39	−0.34	+0.23	−0.29	+0.28	−0.64	+0.30	−0.39	+0.16	−0.59	+0.03	+0.00	−0.01
MAD, all states			0.78	0.42	0.40	0.27	0.40	0.30	0.84	0.38	0.44	0.24	0.78	0.17	0.07	0.04
max |Δ|, all states			2.35	1.93	2.83	2.57	2.95	2.68	2.59	1.87	3.06	2.31	2.27	1.04	0.46	0.18
n, all states			50	50	50	50	50	50	50	50	50	50	50	50	50	50
MSE, valence			−0.70	+0.39	−0.38	+0.22	−0.31	+0.28	−0.76	+0.31	−0.40	+0.17	−0.72	−0.03	+0.00	−0.02
MAD, valence			0.74	0.41	0.40	0.25	0.39	0.28	0.81	0.37	0.42	0.23	0.76	0.14	0.06	0.04
max |Δ|, valence			2.35	1.93	2.83	2.57	2.95	2.68	2.59	1.87	3.06	2.31	2.27	0.85	0.45	0.18
n, valence			42	42	42	42	42	42	42	42	42	42	42	42	42	42

[a] Multireference, [b] inner valence: in the statistics over all states, excluded from the valence ones. — : not available. States with a — are not in the statistics.